

SWRL Rules

Category 1: Stakeholders → Requirements Mapping (12 Rules)

#	Rule Description (First part represent <i>If</i> part and → represent <i>then</i> part)
1	Stakeholder with Regulator role defines requirements with LegalRequirements;RegulatoryCompliance category → NonFunctionalRequirements
2	Stakeholder with ProductOwner role defines requirements with AICostGovernanceRequirements;SystemEndOffLife category → FunctionalRequirements
3	Stakeholder with SecurityEngineer role defines requirements → SecurityRequirements
4	Stakeholder with Operator role defines requirements with MonitoringRequirements;ObservabilityRequirements → FunctionalRequirements
5	Stakeholder with Developer role defines requirements with DataManagementRequirements;ModelLifecycleRequirements → FunctionalRequirements
6	Stakeholder with DataProvider role defines requirements with DataManagementRequirements;TrainingDataQualityRequirements → FunctionalRequirements
7	Stakeholder with ComplianceOfficer role defines requirements with Governance;RegulatoryCompliance → NonFunctionalRequirements
8	Stakeholder with Tester role defines requirements with TestingAndAssurance;Reproducibility → NonFunctionalRequirements
9	Stakeholder with Auditor role defines requirements with AuditAndCompliance;Auditability → SecurityRequirements
10	Stakeholder with Leader role defines requirements with Governance;OrganisationalGovernance → NonFunctionalRequirements
11	Stakeholder with EndUser role defines requirements with Usability → NonFunctionalRequirements
12	Stakeholder with CivilSociety role defines requirements with EthicalDesignRequirements;FairnessAndBiasRequirements → NonFunctionalRequirements

Category 2: Lifecycle → Requirement Mapping (18 Rules)

#	Rule Description (First part represent <i>If</i> part and → represent <i>then</i> part)
13	Requirements with LLM&GenAISpecificRequirements category → toBeImplementedDuring Develop&Build
14	Requirements with ImpactAssessmentRequirements category → hasPhaseValue Plan&Design
15	Requirements with Performance category → hasPhaseValue Deploy&Operate
16	Requirements with DataManagementRequirements category → toBeImplementedDuring Develop&Build
17	Requirements with Fairness&BiasRequirements category → hasPhaseValue Plan&Design
18	Requirements with Learning&AdaptationRequirements category → hasPhaseValue Monitor&Retire
19	Requirements with Governance&Compliance;LegalRequirements category → hasPhaseValue Plan&Design
20	Requirements with ModelLifecycleRequirements category → toBeImplementedDuring Develop&Build
21	Requirements with ExperimentTrackingRequirements;ReproducibilityRequirements category → toBeImplemented Develop&Build
22	Requirements with ExperimentTrackingRequirements category → toBeImplementedDuring Develop&Build
23	Requirements with SystemEndOffLife category → hasPhaseValue Monitor&Retire
24	Requirements with MonitoringRequirements category → hasPhaseValue Deploy&Operate
25	Requirements with HumanAITeamingRequirements category → hasPhaseValue Plan&Design
26	Requirements with AuditAndCompliance category → hasPhaseValue Monitor&Retire
27	Requirements with DataSecurity;ModelSecurity;AccessControl category → toBeImplementedDuring Develop&Build
28	Requirements with PostMarketMonitoringRequirements category → hasPhaseValue Monitor&Retire
29	Requirements with Usability category → hasPhaseValue Deploy&Operate
30	Requirements with IncidentDetectionContainment category → hasPhaseValue DeployAndOperate

Category 3: Requirement → Asset Mapping (30 Rules)

3.1: Functional Requirements → Assets (11 Rules)

#	Rule Description
31	FunctionalRequirements with Governance&Compliance governs ComplianceArtifacts
32	FunctionalRequirements with SoftwareBillOfMaterialsRequirements;ToolAndSandboxingRequirements governs Software&ToolingResources
33	FunctionalRequirements with DataLabelingRequirements governs DataResources
34	FunctionalRequirements with ModelLifecycleRequirements governs Models&Algorithms
35	FunctionalRequirements with SoftwareBillOfMaterialsRequirements governs ComplianceArtifacts
36	FunctionalRequirements with ExperimentTrackingRequirements governs DataResources
37	FunctionalRequirements with ComputeGovernanceRequirements;CloudPlatformRequirements governs InfrastructureResources
38	FunctionalRequirements with WatermarkingAndProvenanceRequirements governs DataResources
39	FunctionalRequirements with DataManagementRequirements governs DataResources
40	FunctionalRequirements with ModelDocumentationRequirements governs Models&Algorithms
41	FunctionalRequirements with ComputeGovernanceRequirements;CloudPlatformRequirements governs InfrastructureResources

3.2: Non-Functional Requirements → Assets (11 Rules)

#	Rule Description
42	NonFunctionalRequirements with Maintainability;Analysability;Modifiability constrains Software&ToolingResources
43	NonFunctionalRequirements with Completeness;TrainingDataGovernanceRequirements constrains DataResources
44	NonFunctionalRequirements with CostManagement constrains InfrastructureResources
45	NonFunctionalRequirements with EnvironmentalSustainability constrains InfrastructureResources
46	NonFunctionalRequirements with Usability;Accessibility constrains Software&ToolingResources

#	Rule Description
47	NonFunctionalRequirements with Security constrains Assets
48	NonFunctionalRequirements with Reliability;FaultTolerance;Recoverability constrains InfrastructureResources
49	NonFunctionalRequirements with Governance&Compliance;OrganisationalGovernance constrains ComplianceArtifacts
50	NonFunctionalRequirements with Scalability constrains InfrastructureResources
51	NonFunctionalRequirements with Portability;Installability;Replaceability constrains InfrastructureResources
52	NonFunctionalRequirements with Performance;Capacity;Scalability constrains InfrastructureResources

3.3: Security Requirements → Assets (08 Rules)

#	Rule Description
53	SecurityRequirements with Logging protects DataResources
54	SecurityRequirements with Privacy protects DataResources
55	SecurityRequirements with AccessControl protects Assets
56	SecurityRequirements with DataSecurity protects DataResources
57	SecurityRequirements with ModelSecurity protects Models&Algorithms
58	SecurityRequirements with AdversarialRobustness protects Models&Algorithms
59	SecurityRequirements with Cloud&InfrastructureSecurity protects InfrastructureResources
60	SecurityRequirements with Privacy protects DataResources

Category 4: Requirement → Requirement Dependency (30 Rules)

#	Rule Description
61	NonFunctionalRequirements (Reliability) dependsOn NonFunctionalRequirements (FaultTolerance)
62	NonFunctionalRequirements (Completeness) dependsOn FunctionalRequirements (DataManagementRequirements)
63	NonFunctionalRequirements (Usability) dependsOn NonFunctionalRequirements (Accessibility)
64	FunctionalRequirements dependsOn SecurityRequirements
65	SecurityRequirements (AdversarialRobustness) dependsOn FunctionalRequirements (InputValidation)
66	SecurityRequirements (NonRepudiation) dependsOn SecurityRequirements (AuditTrail)
67	FunctionalRequirements (SoftwareBillOfMaterialsRequirements) dependsOn SecurityRequirements (VulnerabilityManagement)
68	SecurityRequirements (RAGPipelineSecurity) dependsOn SecurityRequirements (Vector&EmbeddingSecurity)
69	SecurityRequirements (ThirdPartyAIRisk) dependsOn SecurityRequirements (SupplyChainSecurity)
70	FunctionalRequirements (HumanInteractionRequirements) dependsOn NonFunctionalRequirements (Explainability)
71	SecurityRequirements (IncidentDetectionContainment) dependsOn FunctionalRequirements (MonitoringRequirements)
72	SecurityRequirements (AccessControl) dependsOn SecurityRequirements (AccessControl)
73	SecurityRequirements (PromptInjectionDefence) dependsOn FunctionalRequirements (InputSanitization)
74	NonFunctionalRequirements (Portability) dependsOn FunctionalRequirements (Containerization)
75	FunctionalRequirements (Inference&DecisionsRequirements) dependsOn FunctionalRequirements (ModelLifecycleRequirements)
76	SecurityRequirements (DataSecurity) dependsOn SecurityRequirements (AccessControl)
77	NonFunctionalRequirements (Performance) dependsOn NonFunctionalRequirements (Scalability)
78	NonFunctionalRequirements (Interoperability) dependsOn FunctionalRequirements (StandardCompliance)
79	SecurityRequirements (ModelSecurity) dependsOn SecurityRequirements (ModelIntegrityHardening)

#	Rule Description
80	SecurityRequirements (SystemConfidentiality) dependsOn SecurityRequirements (DataSecurity)
81	NonFunctionalRequirements (Explainability) dependsOn FunctionalRequirements (TransparencyRequirements)
82	NonFunctionalRequirements (Recoverability) dependsOn FunctionalRequirements (Backup)
83	FunctionalRequirements (TrainingDataQualityRequirements) dependsOn FunctionalRequirements (DataManagementRequirements)
84	SecurityRequirements (ZeroTrust) dependsOn SecurityRequirements (AccessControl)
85	SecurityRequirements (AgenticAISecurity) dependsOn SecurityRequirements (AccessControl)
86	SecurityRequirements (Privacy) dependsOn SecurityRequirements (DataSecurity)
87	FunctionalRequirements (MonitoringRequirements) dependsOn SecurityRequirements (Logging)
88	FunctionalRequirements (PostMarketMonitoringRequirements) dependsOn SecurityRequirements (IncidentDetectionContainment)
89	FunctionalRequirements (Fairness&BiasRequirements) dependsOn NonFunctionalRequirements (Fairness)
90	SecurityRequirements (Cloud&InfrastructureSecurity) dependsOn SecurityRequirements (InfrastructureSecurity)

Category 5: Classification Rules (18 Rules)

#	Rule Description
91	Requirements containing legal;compliance;regulation;law → NonFunctionalRequirements with LegalRequirements
92	Requirements containing encrypt;access;authentication;privacy;integrity;confidentiality;authorization;audit;security → SecurityRequirements
93	Requirements containing accessibility → NonFunctionalRequirements with Accessibility
94	Requirements containing maintain;modify;test;debug → NonFunctionalRequirements
95	Requirements containing transparency;disclosure;open;visible → FunctionalRequirements
96	Requirements containing latency;throughput;performance;response time;capacity;scalability → NonFunctionalRequirements
97	Requirements containing prompt injection;jailbreak;prompt leakage → SecurityRequirements with PromptInjectionDefence
98	Requirements containing reliability;availability;fault tolerance;recover → NonFunctionalRequirements
99	Requirements containing model integrity;model tampering;model extraction;model stealing → SecurityRequirements
100	Requirements containing fairness;bias;discrimination;equity → NonFunctionalRequirements with Fairness
101	Requirements containing explain;interpret;transparent;understandable → NonFunctionalRequirements
102	Requirements containing prompt injection;jailbreak;prompt leakage → SecurityRequirements
103	Requirements containing privacy;GDPR;personaldata;consent;anonymization → SecurityRequirements with Privacy
104	Requirements containing usability;user friendly;learn;operate → NonFunctionalRequirements
105	Requirements containing shall, not containing encrypt;latency;performance → FunctionalRequirements
106	Requirements containing agent;autonomous;tool invocation;multi-agent → FunctionalRequirements
107	Requirements containing safety;harm;risk;hazard → FunctionalRequirements
108	Requirements containing data governance;data quality;data provenance;data lineage → FunctionalRequirements with DataManagementRequirements

Category 6: Inference Rules (12 Rules)

#	Rule Description (First part represent <i>If</i> part and $\rightarrow$ <i>then</i> part)
109	Requirements with assignee (stakeholder) $\rightarrow$ accountableFor(stakeholder, requirement)
110	Requirements with no priority $\rightarrow$ missingPriority(requirement, true)
111	Assets with no owner and no manager $\rightarrow$ orphanAsset(asset, true)
112	Requirements with multiple assignees $\rightarrow$ hasMultipleAssignees(requirement, true)
113	Requirements with multiple phases $\rightarrow$ hasMultiplePhases(requirement, true)
114	SecurityRequirements with DataSecurity;AccessControl;Privacy;ModelSecurity $\rightarrow$ hasMultiplePhases(requirement, lifecycle)
115	Requirements with assignee, phase, governs/protects/constrains $\rightarrow$ hasFullTraceability(requirement, true)
116	Stakeholder with missing role $\rightarrow$ hasMissingRole(stakeholder, true)
117	Requirements with no phase $\rightarrow$ missingPhaseAssignment(requirement, true)
118	Asset dependency chain ($a1 \rightarrow a2 \rightarrow a3$) $\rightarrow$ assetDependsOnChain(a1, a3)
119	Requirements with no assignee, no phase, no governs/protects/constrains $\rightarrow$ missingTraceability(requirement, Traceability)
120	Asset protected by 4+ security requirements $\rightarrow$ IsCriticalAssetAmong(asset, asset)

Category 7: Validation Rules (08 Rules)

#	Rule Description (First part represent <i>If</i> part and $\rightarrow$ <i>then</i> part)
121	Requirements with no assignee, no phase, no govern/protect/constrain $\rightarrow$ missingTraceability(requirement, true)
122	Requirements with no priority $\rightarrow$ missingPriority(requirement, true)
123	Requirements with no phase $\rightarrow$ missingPhaseAssignment(requirement, true)
124	Assets with security requirements but unprotected $\rightarrow$ unprotectedAsset(asset, true)
125	Requirements unassigned among requirements $\rightarrow$ unassignedRequirement(requirement, true)
126	Stakeholder with missing role $\rightarrow$ hasMissingRole(stakeholder, true)
127	Requirements with missing stakeholder assignee $\rightarrow$ missingStakeholderAssignee(requirement, true)
128	Assets with no governance, protection, or constraint $\rightarrow$ incompleteAssetCoverage(asset, true)